

Yilang Liu

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EDUCATION

Yale University

Ph.D. Candidate in Mechanical Engineering

M.S. & M.Phil in Mechanical Engineering (en route to Ph.D.)

New Haven, CT

Expected Graduation May 2027

Carnegie Mellon University

M.S. in Mechanical Engineering – Research Program

Pittsburgh, PA

May 2022

Chongqing University

B.E. in Mechanical Design, Manufacture and Automation

Chongqing, China

April 2020

University of Cincinnati

B.S. in Mechanical Engineering (Cum Laude)

Cincinnati, OH

April 2020

PUBLICATION

You, Haoxiang, **Liu, Yilang**, Ian Abraham "[Accelerating Visual-Policy Learning through Parallel Differentiable Simulation](#)". Advances in Neural Information Processing Systems (**Spotlight** NeurIPS 2025)

Liu, Yilang, Haoxiang You, and Ian Abraham. "[Sample-Based Hybrid Mode Control: Asymptotically Optimal Switching of Algorithmic and Non-Differentiable Control Modes](#)". IEEE International Conference on Robotics and Automation (ICRA) 2026,

Christian Hughes, **Yilang Liu**, Julia Engdahl, Houston Warren, Darrick Lee, Fabio Ramos, Travis Miles, Ian Abraham "Asymptotically Optimal Ergodic Coverage on Generalized Motion Fields". Robotics: Science and Systems (RSS) 2026

Liu, Yilang, and Haoxiang You. "[Hybrid Mode Sample-Based Control](#)". RSS 2025 Workshop on Whole-Body Control and Bimanual Manipulation: Applications in Humanoids and Beyond, 2025,

Liu, Y., and Barati Farimani, A. "[An Energy-Saving Snake Locomotion Pattern Learned in a Physically Constrained Environment With Online Model-Based Policy Gradient Method](#)." ASME. J. Mechanisms Robotics.

Y. Liu, Y. Chen, H. Li. "[Design an Augmentation Exoskeleton to Enhance lifting Strength](#) " ASME IMECE2020-24658, 2020

RESEARCH EXPERIENCE

Asymptotically Optimal Ergodic Coverage on Generalized Motion Fields

New Haven, CT

Advisor: Prof. Ian Abraham

06/2025-Present

- Developed a flow-adaptive ergodic trajectory optimization framework for robotic exploration in time-varying environments using Maximum Mean Discrepancy (MMD) metrics.
- Derived a theoretical formulation of ergodic coverage on evolving domains, enabling coverage guarantees under dynamic flow fields.
- Designed and implemented a sampling-based trajectory optimization algorithm capable of planning exploration paths in non-convex and flow-constrained environments.
- Demonstrated the approach on multiple robotic platforms, including aerial drones (crazyflies) and Unitree Go2 quadruped robots, validating coverage performance in both simulation and real-world experiments.
- Achieved up to **42% improvement** in exploration performance over information-theoretic baselines in dynamic flow environments.

[Accelerating Visual-Policy Learning through Parallel Differentiable Simulation](#)

New Haven, CT

Advisor: Prof. Ian Abraham

01/2025-06/2025

- Proposed **D.Va (Decoupled Visual-Based Analytical Policy Gradient)**, a novel method for accelerating visual policies by **10x wall-clock time** using parallel differentiable simulation.
- Proposed to decouple visual observations from computation graphs, eliminating differentiating through rendering and achieving training a humanoid running policy in **4 hours** on a single GPU.
- Implemented visual policy learning method from differentiable rendering as baseline comparison using PyTorch3D and proving smoother first-order policy gradient without differentiating through render.
- Benchmarked model-free and model-based visual policy learning approaches and showed **5x higher rewards** in humanoid running tasks against existing algorithms.
- Provided an empirical analysis of analytical policy gradients and highlighted the new opportunities for integrating policy learning with trajectory optimization techniques.

Sample-Based Hybrid Mode Control

Advisor: Prof. Ian Abraham

New Haven, CT

12/2024-04/2025

- Designed a sample-based control strategy for high-dimensional long-horizon tasks using hybrid control theory, achieving complex mode switching behaviors.
- Analyzed theoretical correctness and proved performance guarantees on optimizing mode sequencing.
- Achieved **85% higher reward** in manipulation and **41% in humanoid locomotion** vs. baselines, enabling contact-rich in-hand manipulation.
- Implemented state estimation module for real world Unitree Go2 deployment based on an extended Kalman filter using proprioception information.
- Deployed **real-time torque-level** agile locomotion of a Unitree Go2 quadruped capable of switching from footstand to handstand within sample-based predictive control framework.

Stochastic Hybrid Control for Evaluating and Learning Adaptive Locomotion Skills

Advisor: Prof. Ian Abraham

New Haven, CT

08/2023-04/2024

- Designed an approach of assessing locomotion policy optimality via optimal control principles
- Proposed to integrate arbitrary policies learned offline with dynamic models learned online to determine the local optimality for legged locomotion policies
- Demonstrated efficient policy learning empirically by significantly reducing the wall clock training time while still achieving robust adaptive locomotion skills in hours instead of days

PROFESSIONAL EXPERIENCE

Yale University

Graduate Student Researcher

New Haven, CT

09/2022-Present

- Conduct research in robotics and autonomous systems, developing algorithms for data-driven robotics controls and vision-based policy learning.
- Served as Teaching Fellow for ENAS 773: Introduction to Robotics, Control, and Learning, leading recitations, mentoring student projects, holding office hours, and grading coursework.
- Completed Ph.D. qualifying examination and graduate coursework in robotics, control, and machine learning.

Dexmate Inc.

Robotics Intern

Santa Clara, CA

10/2025-01/2026

- Trained a reinforcement learning policy using Proximal Policy Optimization (PPO) on [Dexmate Vega](#), a mobile robot with dual arm design, guided by the motion planning techniques (RRT). Achieving 100% success rate of grasping cube using 6-DoF dexterous hand in Genesis and Isaac Sim environment.
- Deployed imitation learning algorithm Action Chunking with Transformers (ACT) on the real Dexmate Vega robot with policy trained from bimanual teleoperated dataset. Achieving successful bottle grasping and placing from vision.
- Implemented a Vision-Language-Action (VLA) model $\pi_{0.5}$ for real robot deployment and smoothed up the action chunk transition with Real-Time Action Chunking (RTC) in PyTorch. Achieving smooth action execution with high speed control rate (~20Hz).
- Optimized and streamlined the existing training and deployment framework in LeRobot. Achieving comprehensive log and checkpoint saving on Wandb.

SKILLS

Robotic Platform: Simulation: MuJoCo, Isaac Gym, Genesis, Brax. Hardware: ROS 1, ROS 2, Unitree A1, Unitree Go2
Programming Language: Python, C++